

Amir Mollaei (Amir Hossein Mollaei Khass)

Robotics Ph.D. Student | Perception, Safety, and Active Learning

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Education

Lehigh University

Ph.D. student, AIRLab

Advisor: Prof. Nader Motee (motee@lehigh.edu).

Passed the doctoral general examination.

M.S. in Mechanical Engineering, Lehigh University

B.Sc., Sharif University of Technology

Bethlehem, PA

2023–Present

2025

2023

Research Experience

Graduate Research Assistant | AIRLab, Lehigh University

Research focus: robotic perception, safety-critical control, active scene learning, and 3D Gaussian Splatting (3DGS).

- Developed **Splat-CBF**, combining risk-aware collision avoidance and information acquisition through hard safety and soft perception constraints; designed a **fast 3DGS safety barrier** replacing obstacle-wise constraints with one smooth composite barrier.
- Integrated online 3DGS mapping and operational-space safety filtering for a **Kinova Gen3 7-DoF arm in Isaac Sim**; evaluated Random, Agile-gs, COVER, and FisherRF selectors, with collision-free trials.
- Validated safe active perception on a **physical Ackermann-steered robot**, maintaining a positive safety barrier throughout the reported traversal.
- Developed **TRACE**, a distributed next-best-view protocol coordinating information acquisition while keeping each robot's observations and Gaussian map parameters private.
- Proposed **AGILE-GS**, an anchor-guided fast next-best-view method for active 3DGS, reducing selection latency by **one to two orders of magnitude** on evaluated benchmarks.
- Built an Isaac Sim interface integrating online mapping, safety filtering, and view selection, with manual, keyboard teleoperation, and automatic acquisition modes.

Selected Publications

Active Next-Best-View Optimization for Risk-Averse Path Planning

A. Mollaei Khass, G. Liu, V. Pandey, W. Jiang, B. Lei, K. Daniilidis, and N. Motee.

ICRA 2026. arXiv:2510.06481

Conflict-Aware Active Perception and Control in 3D Gaussian Splatting Fields via Control Barrier Functions

A. Mollaei Khass, A. Cosse, V. Pandey, and N. Motee.

CDC 2026, accepted. arXiv:2605.20566

Splat-CBF: Safe Next-Best-View Control in 3D Gaussian-Splat Maps

A. Mollaei Khass, A. Cosse, and N. Motee.

Under review, ICRA 2027. arXiv:2609.23100

AGILE-GS: Anchor-Guided Fast Next-Best-View Selection for Active 3D Gaussian Splatting

A. Mollaei Khass and N. Motee.

Under review, WACV 2027. arXiv:2609.34176

TRACE: Privacy-Preserving Next-Best-View Selection over Distributed 3D Gaussian-Splat Maps

A. Mollaei Khass, A. Cosse, Q. Sun, and N. Motee.

Under review, ICRA 2027.

Technical Skills

Programming: Python, C++, MATLAB.

Robotics and Simulation: ROS/ROS2, Isaac Sim/Isaac Lab, Habitat-Sim, MuJoCo.

Methods and Tools: PyTorch, 3DGS, control barrier functions, convex optimization, Git, Linux.

Platforms: Kinova Gen3, Ackermann-steered mobile robots, RGB-D cameras.

Teaching Experience

Teaching Assistant, Lehigh University | Convex Optimization; Control Systems.